

Do NFL results follow calendar lattices?

294,379 lattices on every franchise, ten families of patterns, two null models, a hold-out test on later seasons, season cycles and thirteen prediction models, then lattices of my own against the point spread. None survives once each team's schedule is accounted for.

Generated 2026-09-28 · 14,522 team-games (both sides of 7,261 games, ties left out) · 1,000 null histories of each kind per test · data: nflverse games.csv

Lattice families with a league-wide signal after correction

0 of 10

spread-weighted shuffle, $p < 0.005$

Team-family pairs flagged as a team's own lattice

14

14.8 expected from noise alone, out of 320

Teams' best 1999-2012 lattices that held their direction in 2013-2025

39%

125 of 320; 42% expected from noise

Best lattice model, games called correctly

58.1%

lasso logistic; point spread 66.7%

Adding lattice inputs to the point spread

Worse

log loss +0.0063; better in 3 of 22 seasons

Bottom line. No family of lattices beats chance once each team's schedule is accounted for. Under the plain shuffle, which ignores opponents and venues, day-interval repeats and gap + k-th previous rules show weak, diffuse structure; it disappears under the spread-weighted shuffle, so it comes from who played whom and where, not from the calendar. No team has more "own lattices" than noise produces (14 flags against 14.8 expected). Every lattice-based model predicts worse than the point spread, and adding lattice inputs to the spread makes it worse, not better. Part II: None of the 15 lattices with a physical reason survives correction (strongest: division underdog, 52.1% over 1999-2025, corrected $p = 0.59$), and the 25 best of 27,356 machine-found lattices covered 50.2% of 12,541 picks on the seasons after they were found.

What the lab did

- **Ten families of lattices** (section 1): day intervals, two- and three-lookback calendar rules like 28/10, the 10-day / 4th-previous type, game-count repeats, streaks, phases (day number mod n), cycles in game order and in calendar time, and compressibility. 294,379 individual lattices in all, each tested on every team and on all 32 pooled.
- **Two null models**, both keeping every team's exact record in every season: a *shuffle* deals each season's results out in random order; a *spread-weighted shuffle* deals the same wins out again but lets them land more often on games the team was favored in, so it also keeps who played whom and where. A lattice is real only if it beats both.
- **Four kinds of evidence** per family: the strongest single lattice (corrected for the whole family), many weak lattices at once, each team's own lattice (1,000 null histories per team), and a hold-out test: patterns found in 1999-2012 scored on 2013-2025.
- **Calibration and power** (section 8): on pure noise the engine flags about as many teams as chance predicts, and it finds lattices planted in noise.
- **Season cycles** (the 23-year idea) and **thirteen prediction models** trained league-wide, walk-forward, against the point spread. **Part II** (sections 9-12) adds lattices of my own, scored against the spread and registered for the rest of 2026.

1 What was tested

Each family is a whole set of lattices of one kind. A lattice is scored by how far its real record is from what the null model expects, as a z-score (0 = exactly as expected; beyond ± 2 is unusual for one lattice, but with thousands of lattices some will be).

Family	What it asks	Example	Lattices
Day-interval repeats	Does a game repeat (or reverse) the result of the game exactly d days earlier? Every d from 1 to 400 days, plus the long lags 1380, 1764, 1998, 2160, 2760 and 8400 days.	7, 10, 14, 28 days apart	406
Two-lookback rules	When the games d1 and d2 days back had the same result, does the next game repeat it? Every pair of distances up to 63 days.	the 28/10 rule	4,096
Three-lookback rules	Three earlier games at set day distances (up to 42 days) all agree: does the next game follow them?	28, 21 and 7 days	1,747
Gap + k-th previous rules	The game comes g days after the last one (any gap, or exactly 1-21 days) and the last and k-th previous games agree (k = 2-8), in any season or the same season: repeat?	10-day gap, 4th previous	396
Game-count repeats	Same result k games apart, k = 1-40. k = 1 is the classic runs test for streakiness.	4 games apart	40
Streak patterns	Win rate right after each run of the last 1-5 results.	after L-L-L	62
Phase lattices	Win rate on days whose day number leaves remainder a when divided by n, for n = 2-60 (n = 7 is the day of the week). Tested per team only: pooled, every game is one win and one loss.	every 3rd day; Thursdays	1,829
Cycles in game order	Fourier periodogram of each team's season-adjusted W/L sequence: is there a cycle every 2-100 games?	a cycle every 4 games	99
Cycles in calendar time	The same over calendar days: a cycle every 3-400 days, 300 periods.	a cycle every 21 days	300
Compressibility	Lempel-Ziv complexity of the W/L string. Any hidden repeating structure makes a sequence easier to compress.	one number per team	1

The two null models

Shuffled seasons. Every team keeps its exact won-lost record in every season, but the order of the results within the season is random. Anything a lattice finds beyond this is order that the season records alone do not explain.

Spread-weighted shuffle. Same records, but each game's chance of being one of the wins follows the point spread (odds proportional to $e^{b \cdot \text{spread}}$, $b = 0.116$ fitted on how wins fall within team-seasons). Favored games take the wins more often, so the schedule — opponent strength, home and away, rest — is built in. This is the main null.

Every test draws 1,000 histories from each null. The first 1,000 learn what each lattice looks like under the null; a second, independent 1,000 are scored exactly like the real history, so the real data is compared with like. The z-scores use the observed spread of each lattice across null histories, so overlapping cases and small samples are accounted for.

2 League-wide: the summary matrix

One row per family, one column per kind of evidence. Each number is a p-value: the share of noise histories that did at least as well as the real one. Small is interesting; 0.005 is the bar after correcting for ten families (shaded dark), 0.05 is shaded light.

Table 1. p-values by family and test. “Teams flagged” = teams whose own strongest lattice beats noise at $p < 0.05$, real / expected from noise.

Family	Spread-weighted shuffle (main)					Shuffled seasons			
	Strongest lattice	Many weak ones	Teams flagged	Teams p	Held up 2013-25	Strongest	Many weak	Teams	Held up
Day-interval repeats	0.74	0.14	2 / 1.6	0.47	0.40	0.048	0.004	2 / 1.6	0.91
Two-lookback rules	0.82	0.15	1 / 1.6	0.79	0.35	0.10	0.009	1 / 1.6	0.85
Three-lookback rules	0.69	0.76	0 / 1.6	1.00	0.68	0.54	0.39	1 / 1.6	0.38
Gap + k-th previous rules	0.20	0.041	0 / 1.6	1.00	0.72	0.20	0.004	0 / 1.6	0.53
Game-count repeats	0.73	0.82	2 / 1.6	0.48	0.89	0.11	0.32	3 / 1.6	0.75
Streak patterns	0.36	0.71	1 / 1.6	0.81	0.55	0.18	0.36	1 / 1.6	0.18
Phase lattices	—	—	3 / 1.6	0.21	0.88	—	—	2 / 1.6	0.53
Cycles in game order	0.50	0.79	2 / 1.6	0.47	0.92	0.47	0.81	3 / 1.6	0.82
Cycles in calendar time	0.68	0.100	1 / 1.6	0.80	0.57	0.80	0.085	1 / 1.6	0.89
Compressibility	0.63	0.80	2 / 0.7	0.16	0.40	0.53	0.97	2 / 0.8	0.35

“Strongest lattice”: the best of all 32 teams pooled, corrected for every lattice in the family. “Many weak ones”: the average squared z-score of the whole family, which picks up diffuse structure no single lattice shows. “Held up”: each team’s strongest lattice found in 1999-2012, scored on 2013-2025. Phase lattices have no pooled test (pooled, every game is one win and one loss).

Table 2. The strongest league-wide lattices of all families (spread-weighted shuffle). “One-off p” ignores that thousands were tried; “family p” accounts for it. Rates are same-result rates (repeat families) or win rates (streaks); cycles and compressibility show the statistic itself.

Family	Lattice	Real vs expected (games)	z	One-off p	Family p
Gap + k-th previous rules	15-day gap, 5th previous	77.1% vs 55.5% (48)	+3.1	0.003	0.20
Gap + k-th previous rules	15-day gap, 5th previous, same season	81.8% vs 56.7% (33)	+3.0	0.006	0.25
Cycles in calendar time	cycle every 13.7 days	1.391 vs 0.913	+3.0	0.009	0.68
Two-lookback rules	28 & 21 days	60.9% vs 58.4% (3,134)	+2.9	0.002	0.82
Gap + k-th previous rules	15-day gap, 7th previous	75.5% vs 55.0% (49)	+2.9	0.003	0.31
Three-lookback rules	17, 11 & 4 days	26.7% vs 61.7% (15)	-2.9	0.004	0.69
Day-interval repeats	322 days apart	54.4% vs 52.7% (6,228)	+2.8	0.002	0.74
Two-lookback rules	57 & 25 days	92.9% vs 57.3% (14)	+2.8	0.005	0.95
Cycles in game order	cycle every 3.13 games	1.346 vs 0.905	+2.8	0.005	0.50
Cycles in calendar time	cycle every 215 days	1.006 vs 0.681	+2.7	0.010	0.85
Two-lookback rules	53 & 18 days	70.1% vs 59.7% (144)	+2.7	0.015	0.98
Day-interval repeats	310 days apart	70.6% vs 52.6% (51)	+2.7	0.009	0.92

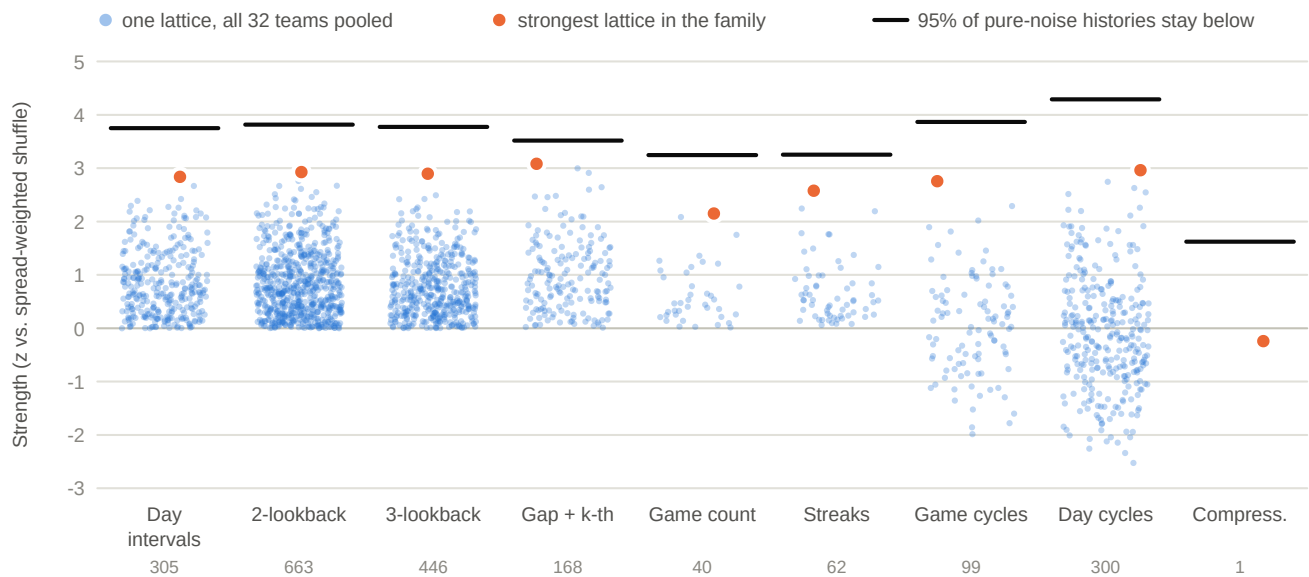


Figure 1. Every league-wide lattice as a dot (2,084 with enough games; count under each family). The black tick is the height that the strongest lattice of a pure-noise history stays below 95% of the time; a real lattice matters only above it. No family’s strongest lattice clears it.

3 Day intervals and the named numbers

The hand analysis singled out intervals of 3, 4, 7, 10, 11, 14, 21 and 28 days and long lags of 1380, 1764, 1998, 2160 and 2760 days. Figure 2 shows every interval from 1 to 400 days at once, league-wide. Pairs of games only exist at some intervals, mostly near whole weeks, so the dots cluster at 7, 14, 21 ... days.

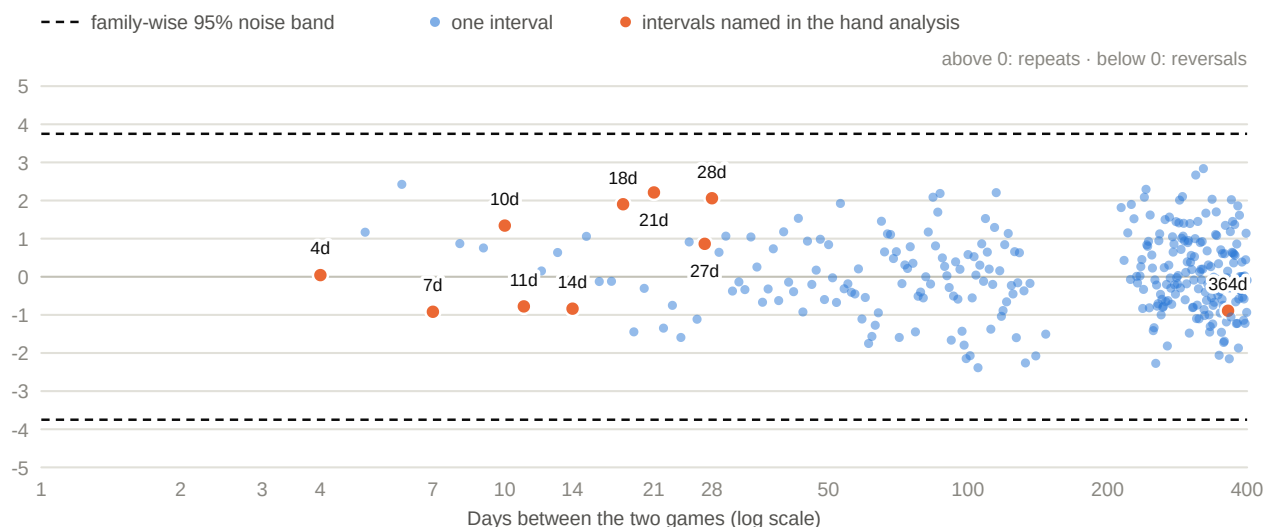


Figure 2. Same-result z-score for each day interval, all 32 teams pooled, against the spread-weighted shuffle. Above zero the two games repeat more often than expected; below zero they reverse more often. The dashed lines are the family-wise noise band for all 406 intervals together: no interval crosses it.

Table 3. The intervals and rules named in the hand analysis. “Real vs expected”: how often the two games had the same result, against what the spread-weighted shuffle expects. p-values are for that one lattice alone, with no correction for the others (the plain shuffle gives similar values). “Too few”: under 8 cases.

Lattice	All 32 teams				Seattle Seahawks			
	Cases	Real vs expected	z	p	Cases	Real vs expected	z	p
Same result 3 days apart	no two games this far apart							
Same result 4 days apart	579	54.4% vs 54.3%	+0.0	0.97	19	52.6% vs 54.4%	-0.2	1.00
Same result 7 days apart	9,119	53.7% vs 54.2%	-0.9	0.35	294	48.3% vs 51.9%	-1.3	0.22
Same result 10 days apart	569	56.9% vs 54.3%	+1.3	0.17	15	80.0% vs 55.0%	+2.0	0.063
Same result 11 days apart	599	53.1% vs 54.7%	-0.8	0.44	23	47.8% vs 53.0%	-0.5	0.70
Same result 14 days apart	8,618	54.1% vs 54.5%	-0.8	0.40	276	44.9% vs 52.7%	-2.7	0.007
Same result 18 days apart	501	58.7% vs 54.5%	+1.9	0.068	20	75.0% vs 52.0%	+2.1	0.037
Same result 21 days apart	8,202	55.6% vs 54.5%	+2.2	0.029	258	51.6% vs 52.7%	-0.4	0.76
Same result 23 days apart	57	49.1% vs 54.0%	-0.7	0.51	3	too few		
Same result 27 days apart	1,037	55.6% vs 54.4%	+0.9	0.39	40	62.5% vs 53.9%	+1.2	0.31
Same result 28 days apart	7,597	55.5% vs 54.4%	+2.1	0.043	238	56.7% vs 52.6%	+1.3	0.19
Same result 161 days apart	no two games this far apart							
Same result 184 days apart	no two games this far apart							
Same result 1380 days apart	608	50.5% vs 51.5%	-0.5	0.59	19	47.4% vs 47.2%	+0.0	1.00
Same result 1764 days apart	4,032	50.9% vs 50.9%	-0.0	1.00	135	59.3% vs 50.2%	+2.2	0.034
Same result 1998 days apart	no two games this far apart							
Same result 2160 days apart	426	51.4% vs 51.1%	+0.1	0.94	16	56.3% vs 51.7%	+0.4	0.81
Same result 2760 days apart	no two games this far apart							
Same result 8400 days apart	1,386	47.9% vs 50.1%	-1.8	0.10	45	53.3% vs 48.2%	+0.7	0.54
28/10 rule: games 28 and 10 days back agree	206	61.2% vs 58.0%	+1.0	0.35	8	100.0% vs 60.6%	+2.4	0.026

Lattice	All 32 teams				Seattle Seahawks			
	Cases	Real vs expected	z	p	Cases	Real vs expected	z	p
10-day gap, 4th previous game	299	60.9% vs 57.7%	+1.2	0.24	10	90.0% vs 58.6%	+2.0	0.031
10-day gap, 4th previous, same season	250	61.6% vs 57.8%	+1.3	0.20	9	100.0% vs 57.9%	+2.6	0.016
11-day gap, 3rd previous ("11/3")	40	57.5% vs 58.7%	-0.2	0.87	2	too few		
21-day gap, 7th previous ("21/7")	2	too few			0	too few		

League-wide, 2 of the 18 named lattices with enough cases reach $p < 0.05$ on their own, against 0.9 expected by luck; the largest deviation is "same result 21 days apart" at $z = +2.2$. Some named numbers never occur as an exact gap between two games of the same team: 3 days, 161 days, 184 days, 1998 days, 2760 days. Three days would mean playing on a Thursday and again that Sunday, and 161 and 184 days reach from September into the off-season.

For Seattle Seahawks, the 28/10 and 10-day/4th-previous rules show small p-values here because they were found by searching these very games. The Seattle analysis corrects for that search, and it leaves them at $p = 0.57$ and 0.56 .

4 Does every team have its own lattice?

If each franchise ran on its own pattern, some teams would stand out in some families far more often than noise allows. Figure 3 gives each team's family-wise p-value in each family: the chance that pure noise produces a lattice as strong as that team's best one. With 32 teams, about 1.6 per family fall below 0.05 by luck alone.

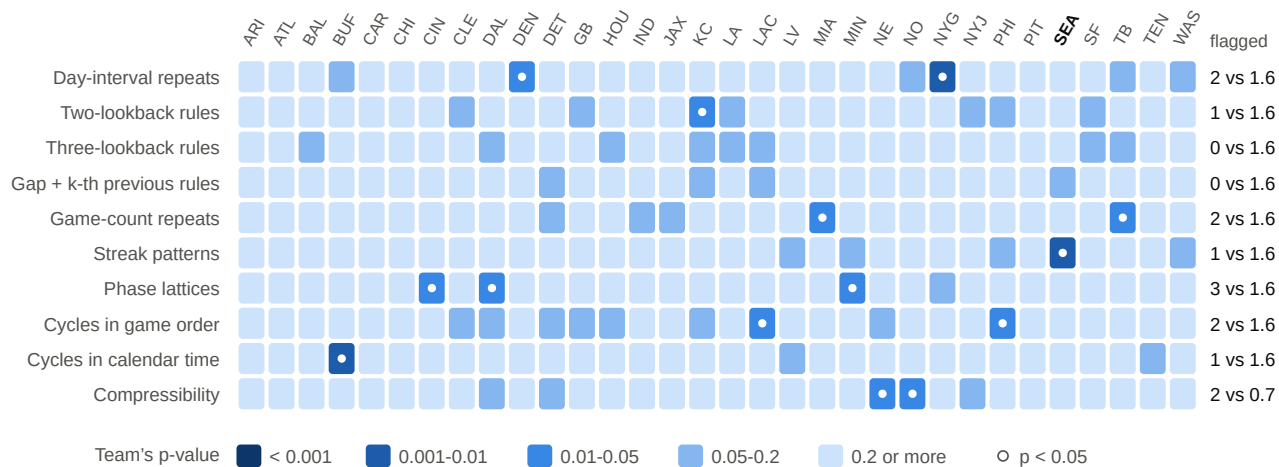


Figure 3. Each cell is one team in one family, against the spread-weighted shuffle; darker = stronger. White dots mark $p < 0.05$. Right: teams flagged in that family versus the number expected from noise. In all: 14 flags against 14.8 expected.

Table 5. Every flagged team-family pair, its strongest lattice, and what that lattice did on the later seasons. “Later z” is the 2013-2025 z-score of the team's strongest 1999-2012 lattice in that family, signed so that + means it kept its direction.

Team	Family	Strongest lattice (1999-2025)	Real vs expected	Team p	Later z
DEN	Day-interval repeats	301 days apart	69.3% vs 53.9% (140)	0.016	+1.5
NYG	Day-interval repeats	1764 days apart	33.6% vs 51.1% (128)	0.003	+2.9
KC	Two-lookback rules	21 & 13 days	16.7% vs 62.0% (12)	0.045	+1.1
MIA	Game-count repeats	39 games apart	42.8% vs 50.6% (407)	0.041	+1.3
TB	Game-count repeats	32 games apart	43.6% vs 51.1% (424)	0.013	+2.6
SEA	Streak patterns	after L-L-W-W	20.7% vs 55.6% (29)	0.003	+1.8
CIN	Phase lattices	day number $\equiv 7 \pmod{51}$	100.0% vs 45.5% (13)	0.014	–
DAL	Phase lattices	day number $\equiv 53 \pmod{55}$	100.0% vs 39.4% (8)	0.027	–
MIN	Phase lattices	day number $\equiv 3 \pmod{31}$	11.1% vs 51.8% (18)	0.026	+1.8
LAC	Cycles in game order	cycle every 18.2 games	9.314 vs 1.130	0.011	–0.7
PHI	Cycles in game order	cycle every 2.67 games	7.479 vs 0.898	0.016	+3.0
BUF	Cycles in calendar time	cycle every 37.9 days	9.633 vs 0.915	0.008	+5.7
NE	Compressibility	LZ complexity	0.905 vs 0.966	0.026	+0.0
NO	Compressibility	LZ complexity	1.007 vs 1.072	0.037	+2.0

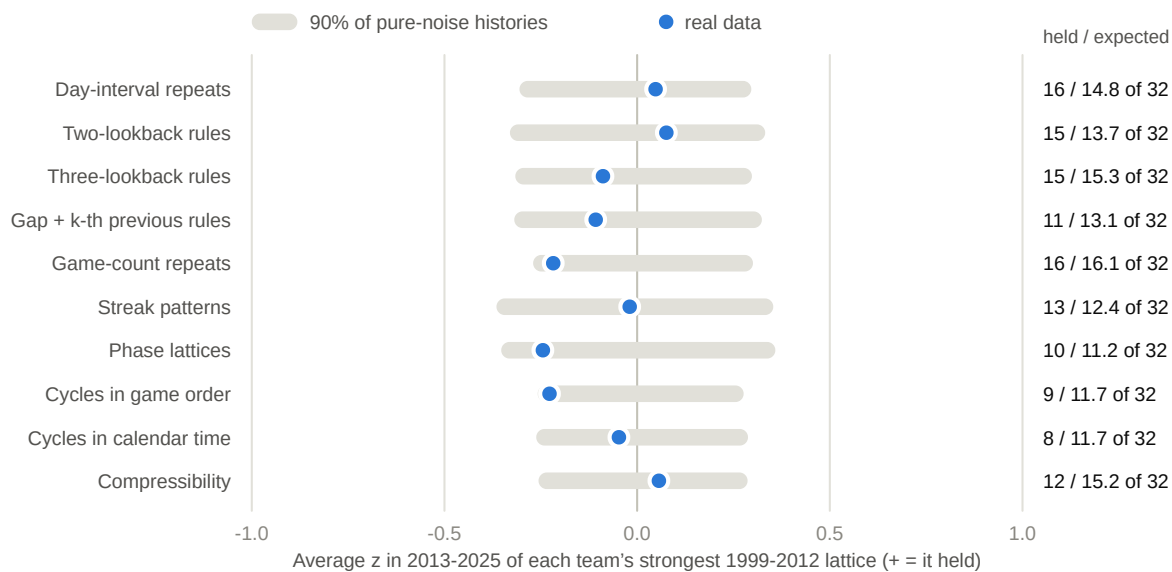


Figure 4. The hold-out test. For every team and family, the strongest lattice found in 1999-2012 was scored on 2013-2025, signed so that + means it kept going the same way. A real pattern shows up as a dot to the right of the gray noise range. Right: how many of the 32 teams' patterns kept their direction.

5 Streaks, phases and cycles

Streak patterns ask the momentum question directly: after W-W or L-L-L, is the next result more likely to repeat? Figure 5 compares the real win rate after every run of up to three results with what the spread-weighted shuffle expects. The expectation is not 50%: teams that just won three straight are usually good teams, and the null keeps that.

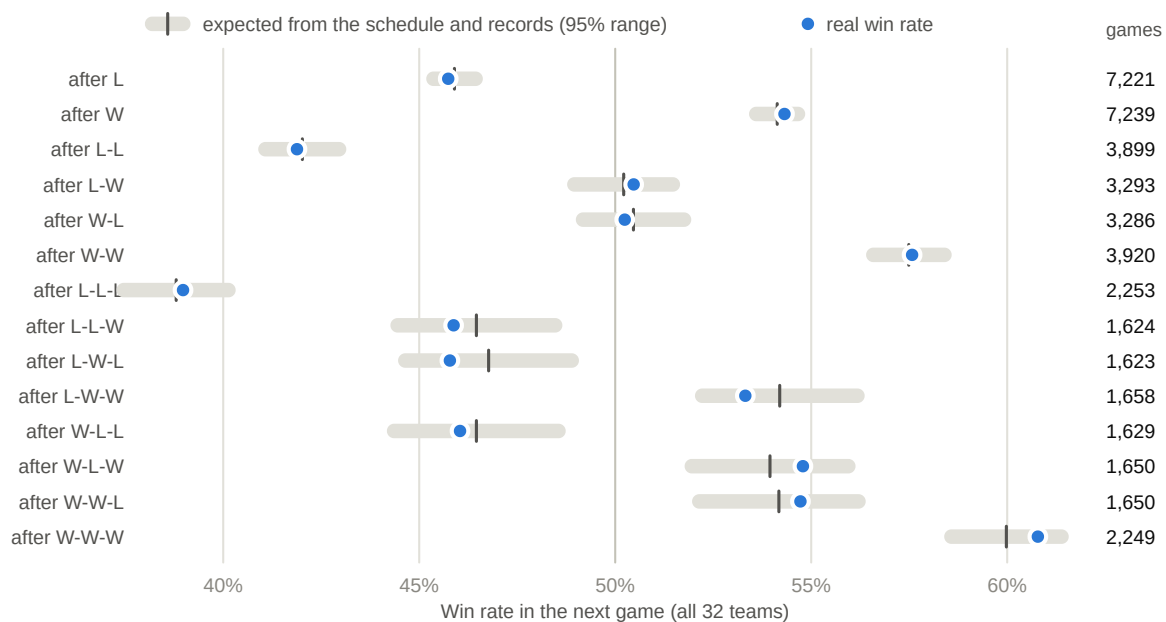


Figure 5. Win rate in the next game after each run of results (oldest first; "W-L" = a win, then a loss), all 32 teams. The gray bar is the 95% range expected from each team's season records and schedule. Streak family, league-wide: strongest pattern $p = 0.36$, whole family $p = 0.71$.

Cycles are the Fourier view of a lattice: a result that repeats every k games, or every P days, puts a peak in the periodogram at that cycle length. Both periodograms follow the noise band closely. The schedule's own rhythm (weekly games, bye weeks, 17-game seasons) is in both the real data and the noise, so it cancels out.

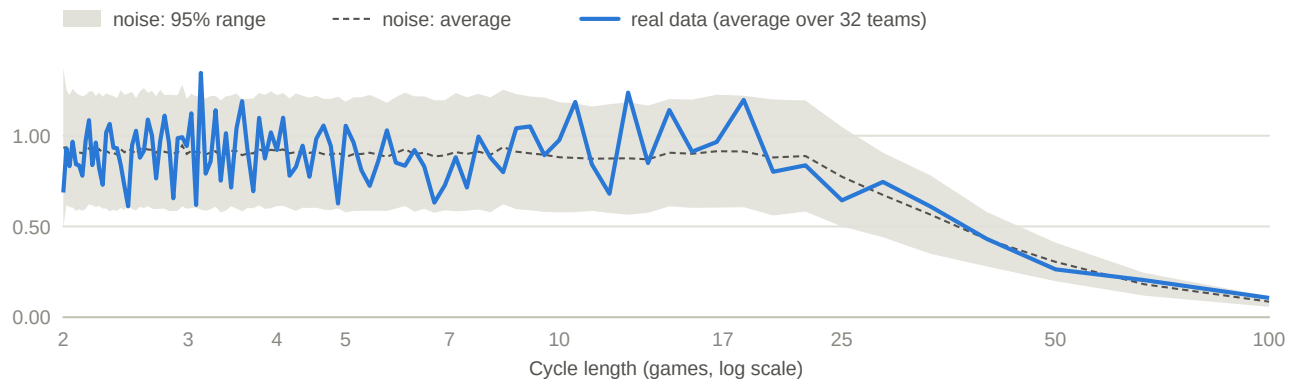


Figure 6a. Cycles in game order, average over the 32 teams. Family p : strongest cycle 0.50, all cycles together 0.79.

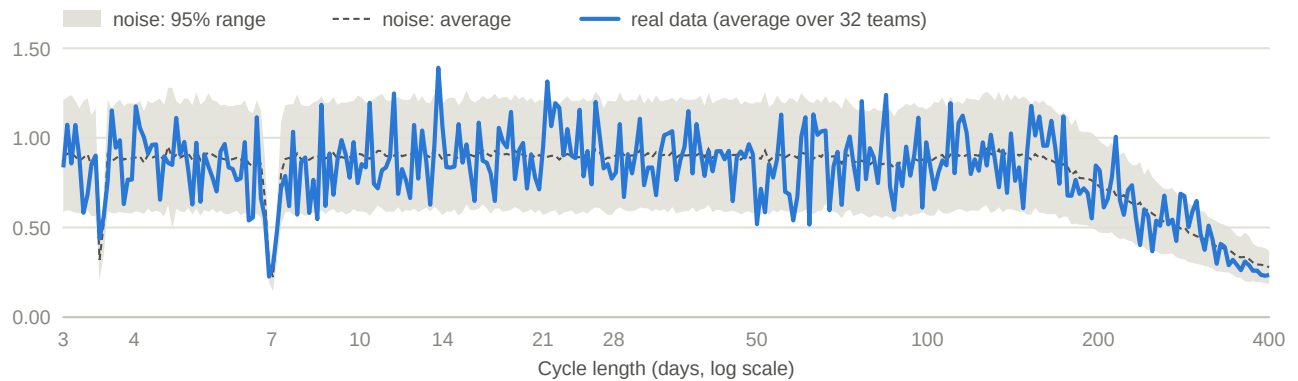


Figure 6b. Cycles in calendar time. Family p : strongest cycle 0.68, all cycles together 0.100. Phase lattices (day number mod n , including weekdays) are tested per team in Figure 3; compressibility, the catch-all test for any repeating structure, gives $p = 0.63$ league-wide.

6 Season cycles: the 23-year batches

The hand analysis proposed 23-year batches ($1999 - 1976 = 23$; $161 = 7 \times 23$; $184 = 8 \times 23$). If seasons echo 23 years later, a team's season win rate should correlate with its win rate 23 seasons before. Figure 7 shows that correlation for every gap from 1 to 26 seasons, all 32 teams pooled, each team measured from its own 1999-2025 average.

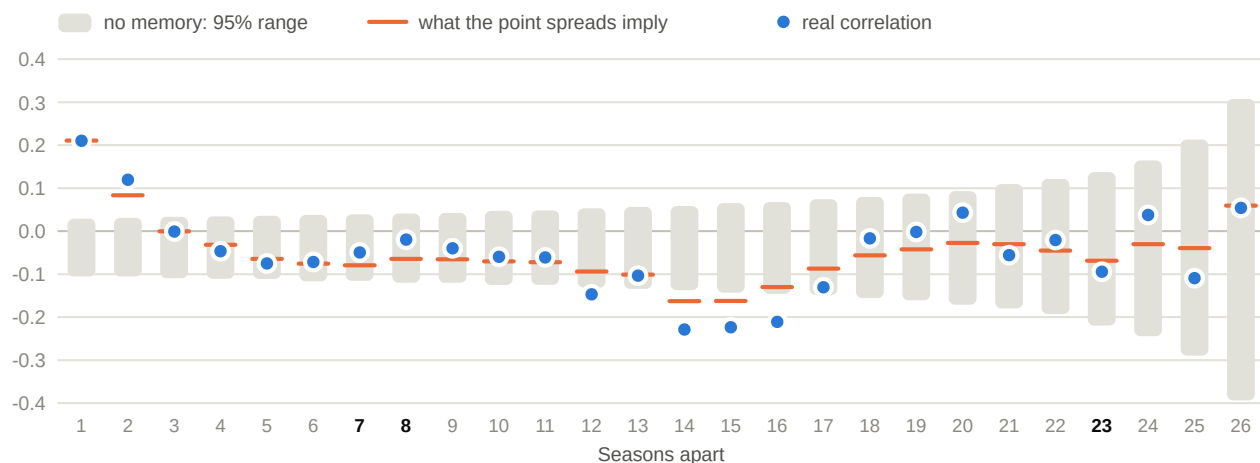


Figure 7. Correlation between season win rates L seasons apart. Gray: the 95% range if seasons had no memory (10,000 random orderings of each team's seasons). Orange: what the point spreads imply, from 1,000 replays of every game. Lags 7, 8 and 23 are in bold.

Seasons apart	Pairs	Correlation	No memory: p	Corrected for 26 lags	Point spreads imply	vs. spreads: p
1	829	0.210	< 0.001	< 0.001	0.211	0.99
2	797	0.120	< 0.001	< 0.001	0.084	0.27
7	637	-0.049	0.79	1.00	-0.080	0.35
8	605	-0.020	0.66	1.00	-0.064	0.23
14	413	-0.229	< 0.001	0.003	-0.163	0.14
15	381	-0.224	< 0.001	0.011	-0.162	0.20
16	349	-0.211	0.002	0.040	-0.130	0.11
23	125	-0.094	0.53	1.00	-0.069	0.76

Good teams stay good for a year or two (one season apart: $r = 0.21$). Franchises also move in long eras: a team far above its own average tends to be below it about 14-16 seasons later (the gaps still flagged after correcting for all 26). All of this is what the point spreads imply too, because the spreads follow the same team quality: against the spreads nothing is left (last column). At 23 seasons the correlation is -0.094, inside the no-memory range ($p = 0.53$), measured on only 125 team pairs because the data starts in 1999.

Table 6. Seattle Seahawks seasons 23 years apart (win rate, playoffs included).

Season	Win rate	23 seasons later	Win rate	Same side of average?
1999	52.9%	2022	50.0%	yes
2000	37.5%	2023	52.9%	yes
2001	56.3%	2024	58.8%	no
2002	43.8%	2025	85.0%	no

Season spread check: real team-seasons stray 16.8% from each team's average (root mean square); replays of every game from the point spread give 14.9% (95% range 14.3%-15.6%). Real seasons are more lopsided than the spreads imply: the market's errors about a team persist within a season. That is team quality the line catches up on, not a calendar lattice, and section 7 shows it cannot be turned into better predictions with lattice inputs.

7 Can any model turn lattices into predictions?

Statistics can miss a pattern that a flexible model would find, so thirteen models were trained on all 32 teams at once, walk-forward: fit on every season before 2004, predict 2004-2008, 2009-2013, 2014-2018, 2019-2025 in turn, never scoring a game the model had seen. The lattice inputs are each team's last 8 results, its results exactly 1-63 days back and at the long named lags, the gaps between games and the weekday. The models range from linear (ridge, lasso) through naive Bayes, a random forest and a neural network to a pattern matcher that looks up what happened after the same recent results, gap and weekday, and per-team models that let every franchise have its own pattern.

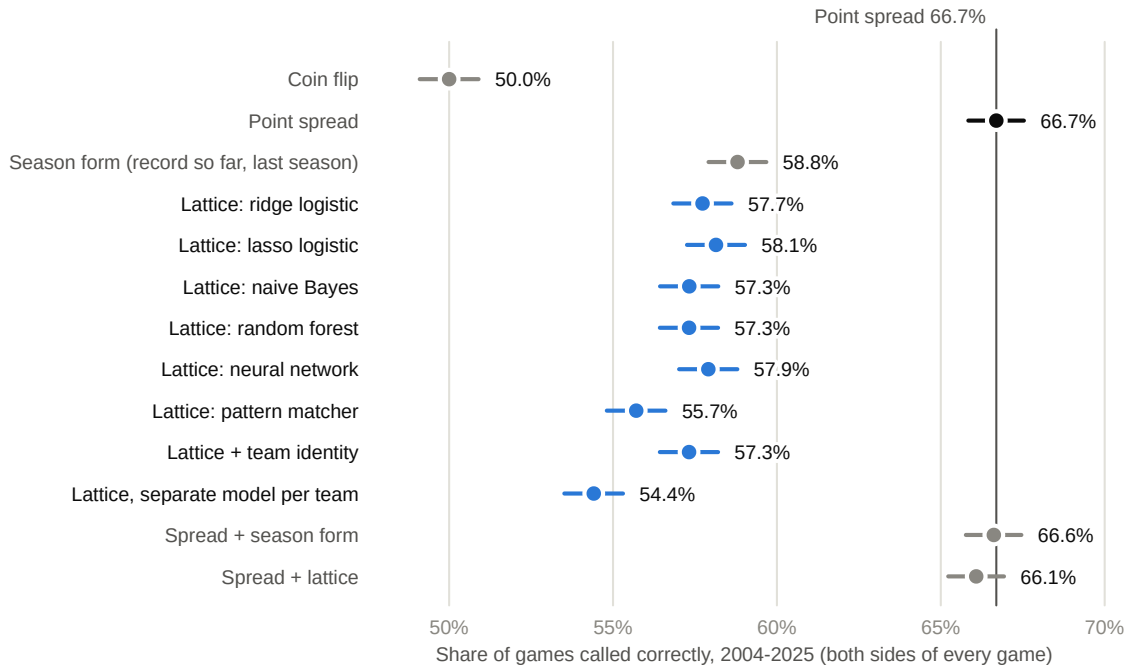


Figure 8. Share of games called correctly, 2004-2025, with 95% ranges. Blue: models that see only lattice inputs (and, for two, the team's identity); black: the point spread; gray: other baselines.

Table 7. Log loss rewards honest probabilities (lower is better; a coin flip scores 0.6931). Δ is the model minus the point spread: positive = worse than the spread. The range is a 95% bootstrap over seasons.

Model	Correct	Log loss	Δ vs spread	95% range	Seasons better
Coin flip	50.0%	0.6931	+0.0843	+0.0762 to +0.0917	0 / 22
Point spread	66.7%	0.6089	benchmark		
Season form (record so far, last season)	58.8%	0.6719	+0.0630	+0.0562 to +0.0691	0 / 22
Lattice: ridge logistic	57.7%	0.6767	+0.0678	+0.0612 to +0.0742	0 / 22
Lattice: lasso logistic	58.1%	0.6768	+0.0679	+0.0612 to +0.0744	0 / 22
Lattice: naive Bayes	57.3%	0.6798	+0.0709	+0.0647 to +0.0772	0 / 22
Lattice: random forest	57.3%	0.6794	+0.0705	+0.0635 to +0.0771	0 / 22
Lattice: neural network	57.9%	0.6777	+0.0688	+0.0623 to +0.0751	0 / 22
Lattice: pattern matcher	55.7%	0.6855	+0.0766	+0.0703 to +0.0831	0 / 22
Lattice + team identity	57.3%	0.6773	+0.0684	+0.0613 to +0.0750	0 / 22
Lattice, separate model per team	54.4%	0.6919	+0.0830	+0.0737 to +0.0929	0 / 22
Spread + season form	66.6%	0.6094	+0.0005	-0.0000 to +0.0010	7 / 22
Spread + lattice	66.1%	0.6152	+0.0063	+0.0041 to +0.0087	3 / 22

Every lattice model lands between 54.4% and 58.1% — better than a coin, because recent results reveal how good a team is — and every one is behind the point spread at 66.7%. Letting each team have its own model makes it worse (54.4%, against 57.7% for one league-wide model): the “own pattern” it learns on early seasons does not carry forward. Adding lattice inputs to the spread raises log loss by 0.0063 (95% range 0.0041 to 0.0087); adding season form instead changes it by 0.0005.

Table 8. Order check: the same lattice models refit on 20 leagues whose seasons were shuffled (records kept). If the order of results carried a lattice, the real history would beat every shuffled one.

Model	Real: correct	Shuffled: average	Shuffled: 5-95%	Real: log loss	Shuffled: average
Lattice: ridge logistic	57.7%	57.1%	56.7%-57.6%	0.6767	0.6784
Lattice: random forest	57.3%	56.8%	56.3%-57.3%	0.6794	0.6802
Lattice: pattern matcher	55.7%	55.1%	54.4%-55.6%	0.6855	0.6895

ridge logistic: 0 of 20 shuffled leagues did at least as well; random forest: 2 of 20 shuffled leagues did at least as well; pattern matcher: 0 of 20 shuffled leagues did at least as well. Where the real history does better, it is by the within-season form that a shuffle destroys (a team that loses its quarterback in week 6 keeps losing), which the point spread already prices in.

8 Calibration: does the engine cry wolf?

A test that flags everything is useless, and so is one that can never flag anything. To check both, one extra history was drawn from each null model — pure noise by construction — and run through the whole engine as if it were the real data, against 300 fresh null histories. A calibrated engine gives it ordinary p-values and flags about 1.6 teams per family.

Table 9. The engine on pure noise: p-values and teams flagged (flagged / expected).

Family	Noise from the spread-weighted shuffle			Noise from the shuffled seasons		
	Strongest	Many weak	Teams	Strongest	Many weak	Teams
Day-interval repeats	0.51	0.22	1 / 1.5	0.68	0.29	1 / 1.5
Two-lookback rules	0.043	0.13	4 / 1.5	0.33	0.69	0 / 1.5
Three-lookback rules	0.92	0.49	1 / 1.5	0.71	0.67	1 / 1.5
Gap + k-th previous rules	0.19	0.033	0 / 1.5	0.037	0.12	1 / 1.5
Game-count repeats	0.78	0.83	1 / 1.5	0.54	0.086	2 / 1.5
Streak patterns	0.15	0.083	2 / 1.5	0.20	0.23	1 / 1.5
Phase lattices	–	–	1 / 1.5	–	–	0 / 1.5
Cycles in game order	0.64	0.82	1 / 1.5	0.82	0.46	3 / 1.5
Cycles in calendar time	0.010	0.75	2 / 1.5	0.98	0.97	1 / 1.5
Compressibility	0.79	0.44	0 / 0.7	0.42	0.70	2 / 0.7

On pure noise, noise from the spread-weighted shuffle got 13 team flags against 14.1 expected, and 2 of 9 strongest-lattice p-values below 0.05; noise from the plain shuffle got 12 team flags against 14.1 expected, and 1 of 9 strongest-lattice p-values below 0.05.

A first version of the engine used null models that re-played games from the point spread without keeping season records. On this check they flagged pure noise as a pattern (real team-seasons are more lopsided than spread-based replays), so they were replaced by the two record-keeping nulls used here.

Would it find a real lattice?

The other half of calibration is power: lattices of known strength were planted in noise histories and the engine was asked to find them (5 plantings per strength, 200 null histories each).

Table 10. Planted lattices. “Found”: the family-wise p-value (league-wide, or the team’s own) was below 0.05. “Top-ranked”: the planted lattice was the strongest of its whole family.

Planted lattice	Planted rate vs expected	Cases	Found	Top-ranked	Median p
10-day gap, 4th previous: repeat (all 32 teams)	64.4% vs 57.9%	293	1 of 5	1 of 5	0.27
10-day gap, 4th previous: repeat (all 32 teams)	71.2% vs 58.0%	293	5 of 5	5 of 5	0.005
10-day gap, 4th previous: repeat (all 32 teams)	79.5% vs 58.2%	293	5 of 5	5 of 5	0.005
Same result 7 days apart (all 32 teams)	55.0% vs 54.2%	9,119	0 of 5	0 of 5	0.38
Same result 7 days apart (all 32 teams)	56.0% vs 54.3%	9,119	1 of 5	3 of 5	0.10
Same result 7 days apart (all 32 teams)	57.9% vs 54.5%	9,119	5 of 5	5 of 5	0.005
Same result 7 days apart (Seattle only)	56.8% vs 52.4%	294	1 of 5	0 of 5	0.28
Same result 7 days apart (Seattle only)	62.2% vs 53.1%	294	2 of 5	3 of 5	0.10
Same result 7 days apart (Seattle only)	71.6% vs 54.9%	294	5 of 5	5 of 5	0.005

League-wide, a 7-day repeat lattice that moves the rate by about 3 points is found 5 times in 5. A 10-day / 4th-previous rule is found once it repeats about 71% of the time instead of 58%, far short of the 90% the hand analysis reported for Seattle. For one team the bar is higher: Seattle’s 7-day interval must move about 17 points. The engine sees lattices of the size the hand analysis described, if they span more than a handful of games, so its silence on the real results is a finding, not a blind spot.

PART II

9 Lattices of my own, against the point spread

Everything so far tested the lattices from the hand analysis and their relatives on wins and losses. This part looks for lattices of my own, and scores them the way that decides whether there is an edge: against the point spread. A lattice picks a side (back a team, or fade it), and the pick wins if that side covers. At the usual -110 odds a pick must win 52.4% of the time just to break even, so 50% means no edge. For reference, over 1999-2025 home teams covered 49.0% of the time and favorites 48.7% (so underdogs 51.3%).

First, lattices with a reason to exist, written down before any of them was scored: rest and byes, the body clock (a 1 pm Eastern kickoff is 10 am for a Pacific team; at night a Pacific team is at its daily peak), travel across time zones, short weeks, rematches, overreaction to blowouts and to cover streaks, and the betting-folklore angles (division and big underdogs, big favorites at night). A full-moon lattice is included as a control that should show nothing. The two hand-found rules are included as bets.

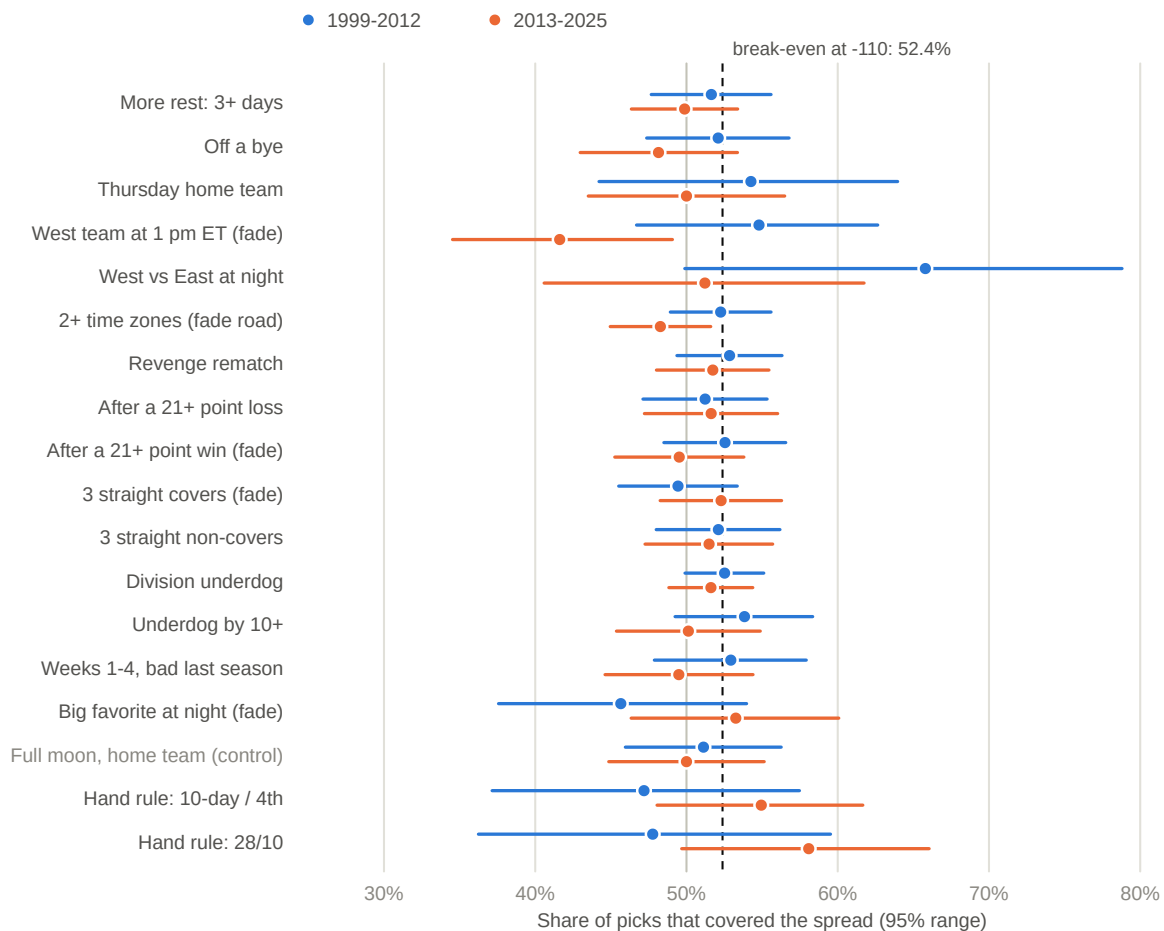


Figure 9. Cover rate of each lattice in 1999-2012 (blue) and 2013-2025 (orange), with 95% ranges. A real edge sits right of the dashed break-even line in both periods.

Table 11. The lattices with a reason, 1999-2025. “Corrected”: Holm over the 18 lattices. “2026”: games played so far this season (through 2026-09-27), not used anywhere else. “Teams differ”: p-value that the cover rate differs by team (32 teams, chi-square). “Seattle”: picks in Seattle games.

Lattice	Picks	Covered	p	Corrected	2013-25	2026	Teams differ	Seattle
More rest: 3+ days More rest than the opponent should help; does the line price it fully?	1,386	50.6%	0.65	1.00	49.9%	5/5	0.82	45/99
Off a bye Teams off a bye against teams that were not.	781	50.3%	0.89	1.00	48.1%	–	0.79	28/59
Thursday home team Thursday games give the visitor a short week and a trip.	318	51.3%	0.69	1.00	50.0%	1/2	0.71	6/19
West team at 1 pm ET (fade) A 1 pm Eastern kickoff is 10 am on a Pacific body clock.	319	47.6%	0.43	1.00	41.6%	2/2	0.89	27/57
West vs East at night At a night kickoff a Pacific team is at its daily peak while an Eastern team is 3 hours later in its day (the circadian finding of Smith et al., 1997).	120	55.8%	0.24	1.00	51.2%	1/1	0.19	17/25
2+ time zones (fade road) Road teams 2+ time zones from home.	1,729	50.3%	0.85	1.00	48.3%	6/7	0.046	161/294
Revenge rematch The loser of the first meeting in a season, in the rematch.	1,479	52.3%	0.077	1.00	51.7%	–	0.62	54/101
After a 21+ point loss After a 21+ point loss the market may overreact.	1,056	51.4%	0.37	1.00	51.6%	6/7	0.024	22/59
After a 21+ point win (fade) After a 21+ point win the market may overreact.	1,111	51.1%	0.47	1.00	49.5%	6/6	0.033	44/81
3 straight covers (fade) Teams that covered 3 straight: the line may chase them.	1,212	50.8%	0.59	1.00	52.3%	–	0.21	48/81
3 straight non-covers Teams that failed to cover 3 straight.	1,106	51.8%	0.24	1.00	51.5%	–	0.43	35/70
Division underdog Division rivals know each other; underdogs may keep it closer.	2,651	52.1%	0.033	0.59	51.6%	8/13	0.61	91/171
Underdog by 10+ Double-digit underdogs: the public backs favorites.	878	52.1%	0.24	1.00	50.1%	0/2	0.79	25/60
Weeks 1-4, bad last season In weeks 1-4 the line may lean too hard on last season.	774	51.2%	0.54	1.00	49.5%	12/20	0.51	23/52
Big favorite at night (fade) Big favorites in night games draw public money.	337	50.1%	1.00	1.00	53.3%	–	0.94	7/20
Full moon, home team (control) Control: nothing should happen. Home teams within 1.5 days of a full moon.	718	50.6%	0.79	1.00	50.0%	7/11	0.73	23/45
Hand rule: 10-day / 4th The hand-found 10-day / 4th-previous rule as a bet: after 10 days, back a team whose last and 4th-previous games were wins, fade one whose were losses.	291	52.6%	0.41	1.00	55.0%	1/1	0.96	7/16
Hand rule: 28/10 The hand-found 28/10 rule as a bet: back a team that won the games 28 and 10 days before, fade one that lost both.	203	54.7%	0.21	1.00	58.1%	–	0.80	6/11

None survives the correction. The strongest, division underdog, covered 52.1% of 2,651 picks (corrected p = 0.59) and 51.6% in 2013-2025, below break-even. 1 of 15 beat the break-even line on 2013-2025, about what luck gives when the true rate is 50-51%. The moon control covered 50.6%. The hand rules as bets: 10-day / 4th 153/291

(52.6%, $p = 0.41$); 28/10 111/203 (54.7%, $p = 0.21$), on every team since 1999. Do the lattices work for some franchises and not others? 3 of 18 show a team-to-team spread with $p < 0.05$ (0.9 expected by chance; the smallest, $p = 0.024$, is 0.43 after correcting for 18), so none belongs to particular franchises either. In Seattle Seahawks games the samples are small: the most extreme, after a 21+ point loss, is 22/59 ($p = 0.067$), one of 15 checked.

10 What the market did not expect

The spread is the market's best guess of the margin; the surprise is the real margin minus the spread. If any lattice drove results beyond what the market knows, surprises would echo: a team that beat the spread would beat it again a set number of games or days later. Figure 10 correlates each team's surprises at every interval from 1 to 8 games and at 17 day intervals, against 2,000 histories where each game's surprise had its sign flipped at random.

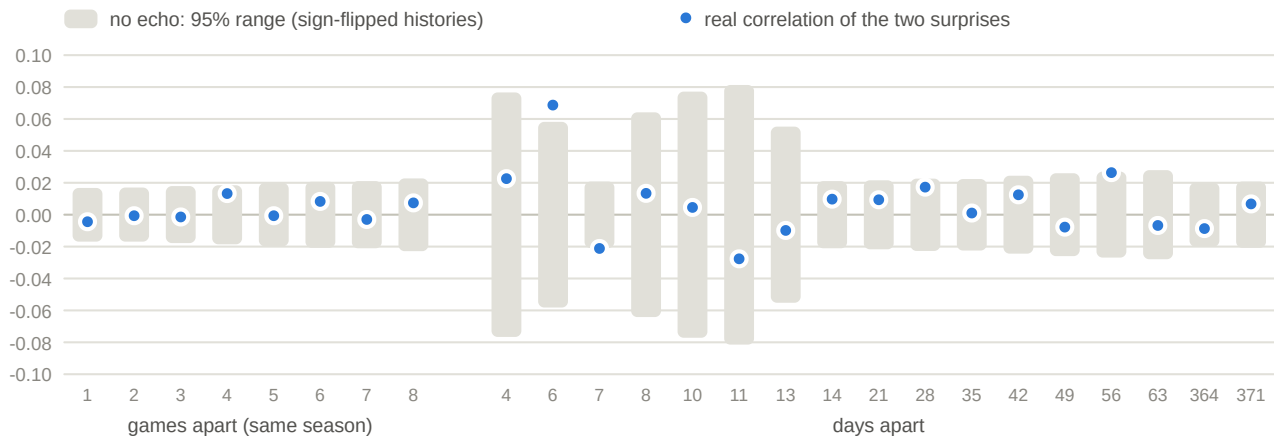


Figure 10. Correlation between a team's spread surprises at the given interval, all teams 1999-2025. Gray: the range that surprises with no echo produce 95% of the time. One game apart: $r = -0.004$. The most extreme interval, 6 days apart ($r = 0.069$, 1,223 pairs), has a family-wise p of 0.30.

Surprises do not echo: beating the spread one week says nothing about the next week, or about the game 7, 14 or 28 days later. This is the cleanest test of the lattice idea, because it removes everything the market already prices in. If a calendar rhythm moved results, this is where it would show.

11 A machine search for lattices

Last, I let the computer look. A grammar of 93 yes/no facts known before kickoff (home or road, the spread, rest and rest difference, weekday, kickoff time, time zones and body clock, month, week, previous results and covers of both teams, streaks, results exactly 7, 14, 21 and 28 days before, the 10-day gap and 4th-previous pattern, the 28/10 pattern, the last meeting with the opponent, this and last season's record, the moon) combined into every lattice of one or two facts and, for the most promising pairs, three: 27,356 lattices, of which 9,860 made at least 150 picks on 1999-2012. Each was scored as a back or a fade, whichever did better.

The best reached $z = 3.97$ (438 of 766 picks). That sounds like $p < 0.001$, but the same search run on 1,000 histories where every game's cover was a coin flip finds a best lattice of $z = 3.93$ on average (95% of the time below 4.50). Family-wise $p = 0.42$: the best lattice is exactly as good as the best lattice in noise.

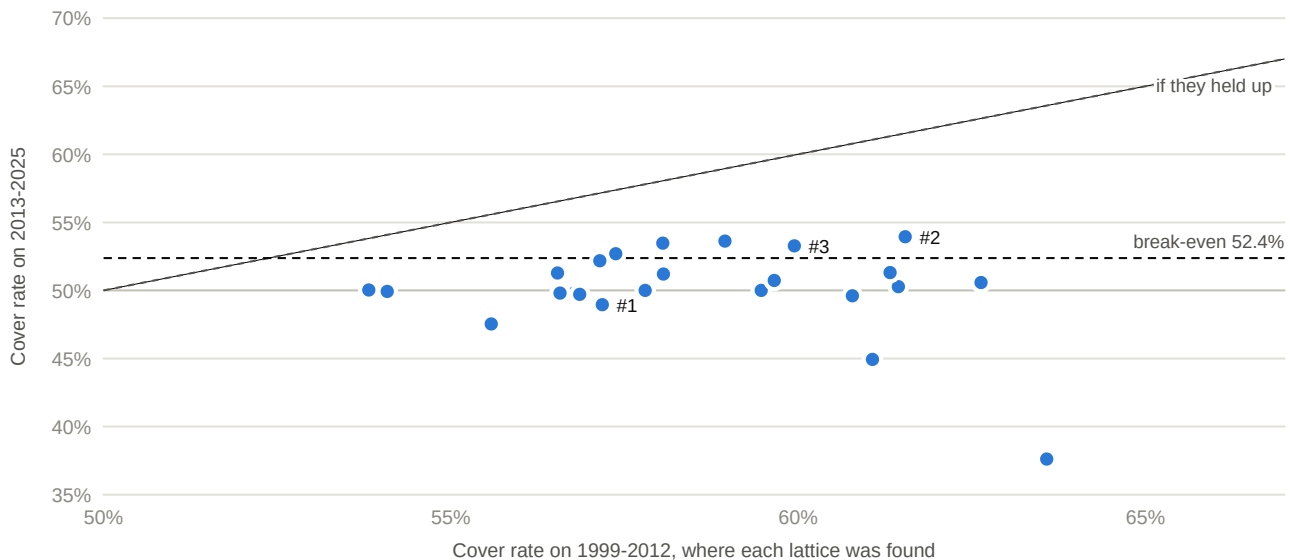


Figure 11. The 25 best machine-found lattices (duplicates that pick the same games removed): cover rate where they were found against cover rate on the 2013-2025 seasons. Real lattices would sit near the diagonal; these fall back to about 50%. Together they covered 50.2% of 12,541 later picks; 13 of 25 stayed above 50% and 5 above break-even.

Table 12. The ten best machine-found lattices.

#	Lattice	Found on 1999-2012	Family p	2013-2025	2026
1	Back the side that is: Sunday game + team lost its last game + lost the last meeting with this opponent	438/766 (57.2%)	0.42	328/670 (49.0%)	6/8
2	Fade the side that is: Central-time team + opponent lost its last game + won the last meeting with this opponent	160/260 (61.5%)	0.72	130/241 (53.9%)	2/3
3	Fade the side that is: home team + division game + November game	208/347 (59.9%)	0.74	138/259 (53.3%)	–
4	Back the side that is: opponent failed to cover its last game + lost the game exactly 7 days before + lost the last meeting with this opponent	172/283 (60.8%)	0.82	127/256 (49.6%)	1/3
5	Fade the side that is: opponent lost its last game + lost the game exactly 21 days before + won the last meeting with this opponent	160/262 (61.1%)	0.85	102/227 (44.9%)	–
6	Back the side that is: team failed to cover its last game + lost the game exactly 7 days before + lost the last meeting with this opponent	281/484 (58.1%)	0.90	212/414 (51.2%)	3/4
7	Back the side that is: team lost its last game + won the game exactly 14 days before + lost the last meeting with this opponent	145/236 (61.4%)	0.91	89/177 (50.3%)	1/1
8	Fade the side that is: home team + kickoff before 2 pm ET + won the last meeting with this opponent	274/472 (58.1%)	0.92	208/389 (53.5%)	2/3
9	Back the side that is: Eastern-time team + opponent failed to cover its last game + lost to this opponent earlier this season	119/190 (62.6%)	0.93	86/170 (50.6%)	–
10	Back the side that is: team lost its last game + lost the game exactly 28 days before + lost the last meeting with this opponent	201/338 (59.5%)	0.93	130/260 (50.0%)	–

12 Registered for the rest of 2026

A lattice found in old games can only be tested fairly on games that have not been played. So every lattice in this part — the 18 with a reason and the top 3 from the machine search — is now written into `ledger.json` with today's date (2026-09-28), next to the two hand-found Seattle rules. `node cli.js ledger` scores each one only on games after that date. These are the picks for the games that already have a line:

Date	Game	Pick against the spread (lattices)
10-01	PIT at CLE	CLE +2.5 (Thursday home team, Division underdog, Weeks 1-4, bad last season)
10-04	ARI at NYG	NYG -1.5 (West team at 1 pm ET (fade), 2+ time zones (fade road))
10-04	DAL at HOU	HOU -2.5 (3 straight non-covers) DAL +2.5 (Weeks 1-4, bad last season)
10-04	DET at CAR	CAR +3 (Weeks 1-4, bad last season)
10-04	GB at TB	GB -3 (More rest: 3+ days, After a 21+ point loss) TB +3 (Weeks 1-4, bad last season, Hand rule: 10-day / 4th)
10-04	JAX at CIN	CIN -2.5 (After a 21+ point win (fade), Weeks 1-4, bad last season)
10-04	KC at LV	LV +5.5 (2+ time zones (fade road), Division underdog) KC -5.5 (3 straight covers (fade))
10-04	LAC at SEA	LAC +6.5 (3 straight non-covers)
10-04	MIA at MIN	MIA +10.5 (3 straight covers (fade), 3 straight non-covers, Underdog by 10+, Weeks 1-4, bad last season)
10-04	NE at BUF	NE +7 (After a 21+ point loss, 3 straight covers (fade), Division underdog, Machine lattice #1)
10-04	NYJ at CHI	NYJ +3 (Weeks 1-4, bad last season)
10-04	TEN at BAL	TEN +11.5 (Underdog by 10+)
10-05	ATL at NO	ATL +2.5 (More rest: 3+ days, Division underdog) NO -2.5 (After a 21+ point win (fade))

None of these lattices showed an edge on 1999-2025, so the expected result is about 50%, and the picks should not be bet. The ledger is the honest way to find out if that is wrong, but it is slow: to show that a lattice winning 55% of its picks really beats the 52.4% break-even (80% power) takes about 2,243 picks, and a typical lattice here makes about 38 picks a season. Showing 55% against a plain coin flip takes about 617.

13 What it means

In plain terms: across 294,379 lattices on all 32 teams since 1999, the results look like what you get when each team's season record is dealt out over its schedule at random, with a bit more of the wins going to the games the betting line favored. Intervals like 7, 10, 14 and 28 days, rules like 28/10 and 10-day/4th-previous, the long numbers (1380, 1764, 1998, 2160, 2760) and the 23-year batches all sit where chance puts them. My own lattices do no better against the point spread: none of the 15 lattices with a physical reason survives correction (strongest: division underdog, 52.1% over 1999-2025, corrected $p = 0.59$), and the 25 best of 27,356 machine-found lattices covered 50.2% of 12,541 picks on the seasons after they were found.

- **Why hand-found lattices look convincing.** Searching many intervals and rules on one team always turns up some that were right 8 or 9 times in 10 by luck. That is why every family here is judged by its strongest lattice against the strongest lattices of noise histories, and by whether it holds up on later seasons. Patterns found that way fail on fresh seasons and on other teams, which is what happened to 28/10 and 10-day/4th-previous in the Seattle analysis.
- **The 3-4-7-10-11-14 numbers are the schedule.** NFL games fall on a weekly grid (Sunday, Monday, Thursday, the occasional Saturday), so the gaps between games cluster at a handful of values: 3, 4, 6, 7, 8, 10, 11, 13 and 14 days. They are real intervals, but they carry no information about results.
- **What does carry information** is ordinary team quality: good teams keep winning within a season and for a year or two, franchises have long eras, and the betting line captures nearly all of it. That is why the point spread calls 66.7% of games and no lattice model beats it.
- **What would change this verdict:** a lattice written down before the games it predicts, beating the point spread on those games over a few seasons. The forward ledger (`node cli.js ledger`) is set up to score exactly that: it now holds 23 lattices dated 2026-09-28, yours and mine.

Reproduce

```
node cli.js lab          # this report: results/lab.json + results/lattice-lab.pdf (~12 min)
node cli.js lab --quick  # fewer null histories (~3 min)
node cli.js lab --parts own,power # recompute only some parts
node cli.js lab-pdf      # rebuild the PDF from results/lab.json
node cli.js ledger       # score the registered lattices on games after their date
```

Method notes

z-score of a lattice

For repeat and streak families, $(\text{hits} - n \cdot \mu) / \sqrt{(n \cdot v)}$, where μ is the lattice's average rate across null histories and v its observed per-case variance there (so overlap between cases is counted). For cycles and compressibility, $(\text{value} - \text{null mean}) / \text{null sd}$. Lattices with fewer than 8 cases are skipped.

Family-wise p

The share of null histories whose strongest lattice in the family is at least as strong as the real strongest one. It answers "how often does noise produce a best lattice this good?" and needs no further correction within the family.

Teams flagged

Each team gets its own family-wise p from 1,000 null histories; the count with $p < 0.05$ is compared with the same count in every null history (each ranked against the others).

Hold-out

Lattices are ranked on 1999-2012 and scored on 2013-2025. The z-score in the later half is signed by the direction found earlier, so + means the pattern continued. The same pick-then-score procedure is run on every null history.

Spread-weighted shuffle

Within each team-season, W wins are placed on the games with probability proportional to the product of $e^{b \cdot \text{spread}}$ over the chosen games (a conditional Bernoulli draw, sampled exactly by dynamic programming). b is the conditional-logit slope, 0.116; across all games the spread's slope is 0.141.

Against the spread

A lattice picks one side of a game: the side that fits its conditions (or the other side, for a fade). Games where both sides fit, or where the point spread is missing, are skipped; pushes are not counted. A pick wins when its side covers (final margin beats the spread). The mechanism lattices were written down before any of them was scored and are corrected together (Holm). The machine search's family-wise p is the share of sign-flipped histories (each game's cover outcome flipped with probability 1/2) whose best lattice is at least as strong.

Machine learning

Penalties are chosen on the last three training seasons, then refit on all training seasons. Ridge and lasso are exact logistic fits (Newton and FISTA); the forest has 100 trees, depth ≤ 8 , leaves of ≥ 40 games; the network is the Seattle analysis's (16 tanh units, 3 seeds, early stopping).

Data: nflverse `games.csv` (Lee Sharpe), every NFL game since 1999; relocated franchises merged (OAK $\rightarrow$ LV, SD $\rightarrow$ LAC, STL $\rightarrow$ LA).